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GASTRORETENTIVE SUPERPOROUS HYDROGEL IN PEPTIC ULCER THERAPY: A COMPREHENSIVE REVIEW

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ABSTRACT:

Peptic Ulcer Disease (PUD) results from the weakening of the protective mechanism of the stomach and duodenum lining. Normally, the mucosal protection is provided by the mucous bicarbonate secretion and prostaglandins, although these two fail to be sufficient against strong attacking agents such as acid, pepsin, Helicobacter pylori bacterial infection, NSAIDs intake, alcohol, and cigarette smoking. Oral therapy treatment is common but is associated with numerous disadvantages. The latter include low drug concentration in the ulcer area and mucus, very short duration in the stomach, chemical instability of some drugs in the acidic stomach environment, and growing antibiotic resistance. One such material is a particularly notable type of hydrogel that is becoming increasingly significant in this field – superporous hydrogels and their hybrid forms. The properties that make this form especially impressive are that it possesses both high mechanical strength and high water content and has very high porosity and a flexible three-dimensional structure. It develops rapidly to a volume sufficient enough to inhibit stomach emptying and hence stay inside the stomach for 10-12 hours while continuously releasing anti-biotics and anti-ulcer agents. Experiments on animals have demonstrated much higher protection against ulcers with this approach, hence indicating the enormous potential of superporous hydrogels.

Keywords: *Pancreatic cancer, Tumour microenvironment, Genetic mutations, Targeted therapy*

Introduction

Peptic ulcers are defined as conditions characterized by the occurrence of extensive ulceration of the inner lining of the stomach, mucosa, and/or duodenum up to the muscular layer due to the production of gastric acid in the surroundings **(Malik et al., 2025)**.Chronic *Helicobacter pylori (Hp)* infection and NSAID usage, are the two most common etiological precursors **(Almadi et al., 2024; Vakil, 2024)**. Although most peptic ulcers are asymptomatic at first, symptoms can vary from mild dyspepsia to problems such gastric outlet obstruction, GI (gastrointestinal) bleeding, and perforation **(Ocasio Quinones & Woolf, 2026)**.

Tobacco consumption (including smoking) is also an important provoking factor in ulcerogenesis and ulcer healing. Alcohol consumption can have similar effects too, although to a lesser extent **(Liu et al., 2020)**.

The damage to the mucosal lining during the development of a peptic ulcer is depicted in **Figure 1**.

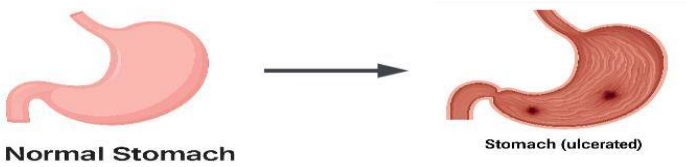


Figure 1: Visual representation of normal stomach mucosa and ulcerated stomach indicating tissue erosion

Etiology and pathophysiology of peptic ulcer

Under normal circumstances, the mucus-bicarbonate barrier, neutral pH, and ongoing epithelial cell renewal preserve the integrity of the duodenal and gastric mucosa **(Almadi et al., 2024)**. NSAID use and Helicobacter pylori infection are the two major causes of peptic ulcers. H. Pylori colonise the gastric mucosa causing inflammation, epithelial destruction and changes in acid secretion (hypo- or hyperchlorhydria) **(Tran et al., 2024)**.This results in release of cytokines and stimulation of the parietal cells along with inhibition of somatostatin, with increased release of histamine and gastrin, which leads to hypersecretion of acid and pepsin. However, mucosal atrophy can develop, resulting in a loss of acid secretion. NSAIDs play a direct role in damage to the gastric mucosa through blockade of the enzyme COX-1, resulting in a decreased production of prostaglandins. The decrease in prostaglandin production causes a decreased mucus production, bicarbonate secretion, and blood flow **(Ko & Lee, 2025)**.

Therapeutic Approaches for Peptic Ulcer

The treatment of peptic ulcer disease includes two types of antisecretory agents, H2-receptor antagonists and proton pump inhibitors . The PPIs have replaced H2 receptor blockers mainly due to better healing and efficacy **(Begg et al., 2023; Meng et al., 2023).** PPIs relieve symptoms and encourage recovery by preventing the stomach from producing acid **(Salman et al., 2025)**. The first line treatment of *H. pylori* induced peptic ulcer is a triple regimen of two antibiotics and a proton pump inhibitor **(Jung et al., 2021)**. PPIs and antibiotics combine to eradicate *H. pylori* **(Shah et al., 2021)**.Quadruple therapy using bismuth and several antibiotics is performed if first-line treatment is unsuccessful **(Aumpan et al., 2023)**.

Limitations of conventional dosage forms

Standard forms of medication in the form of tablets, capsules and suspensions are still widely used in the treatment of peptic ulceration. This also applies to that associated with *Helicobacter pylori* **(Raja, 2025)**. However, these types of formulations come with several significant drawbacks that impact the effectiveness of treatment and patient results. A major issue is the brief gastrointestinal residence duration of standard oral dosage forms. These medications generally move through the stomach quickly, which restricts the time that the drug has contact with the gastric mucosa and the ulcer area **(Gupta et al., 2023, 2025)**. Furthermore, many of the active pharmacological compounds used in the treatment of peptic ulcers such as some antibiotics and proton pump inhibitors are unstable in the acidic environment of the stomach resulting in early degradation and lowered bioavailability **(Andrawes et al., 2025; Iman & mirfakhraee, 2024)**.A significant drawback is the insufficient concentration of medication at the ulcer location. As *H. pylori* exists deep within the gastric mucosal layer, drugs administered through standard methods frequently do not achieve therapeutic levels in these areas. This leads to partial eradication and ongoing infection. Insufficient drug retention and absorption result in sub-therapeutic levels, promoting the emergence of resistant *H. pylori* strains. The effectiveness of first-line antibiotics like amoxicillin, clarithromycin, and metronidazole has deteriorated worldwide due to improper dosing and ineffective delivery **(Grosso & de-Paz, 2022)**.

Concept of novel drug delivery systems

Treatment for stomach ulcers and *H. pylori* is inefficient due to the difficulties described above. Many antibiotics do not adequately reach the target site of action, resulting in poor in vivo efficacy **(Chitas et al., 2024; Iman & mirfakhraee, 2024)**. *H. pylori* is situated in the innermost layer of the stomach lining, where the therapeutic concentration, half-life (t½), and bioavailability of drugs are constrained. The medicine should be given at certain times to ensure that the plasma falls in accordance with the level of the drug

(Gupta et al., 2023). The controlled drug delivery system controls drug bioavailability, and it also improves the efficacy of a medication. It can control drug delivery in one way. GRDDS is one of a number of targeted delivery methods. The system gives a slow delivery of the drug and attempts to retain the formulation in the stomach for a prolonged period. Furthermore, GRDDS exerts a localized effect on the gastric mucosa by administering a concentrated amount of drug to the intended site **(Mokashi et al., 2024; Omidian, 2025; R. I. Patel et al., 2025)**.

Gastroretentive drug delivery systems (GRDDS)

Oral administration remains popular due to patient comfort and convenience, despite ongoing advancements in drug delivery methods **(Lou et al., 2023; Vllasaliu, 2025)**. These medication delivery systems are known as gastroretentive drug delivery systems (GRDDS). GRDDS is ideal for medications that are absorbed in the stomach, such as (ranitidine and metformin),or poorly soluble at alkaline pH (*e.g.* furosemide and diazepam) **(Vinchurkar et al., 2022; Vrettos et al., 2021)**.

Hydrogels

Hydrogels, defined by their distinctive three-dimensional structures composed of water-attracting polymers, have become fundamental to the progress of biomaterial science **(Kesharwani et al., 2021; D. K. Patel et al., 2024)**. These networks, which can absorb and hold large amounts of water, are notable for their extraordinary capacity to expand without disintegration, preserving their structural stability through chemical or physical cross-linking processes **(D. K. Patel et al., 2024; Rybicki et al., 2025)**. Superporous hydrogels (SPHs) are ideal for use in gastro-retentive drug delivery systems due to their large interconnected pore-network which facilitates rapid swelling (generally within minutes) and large volume expansion. These features are especially advantageous in the treatment of peptic ulcers, as the fact that the drug is released in a controlled manner and it resides longer in the stomach increase its effectiveness **(Issarachot et al., 2023; Pawar et al., 2024)**.

Superporous Hydrogel

Superporous hydrogel (SPH) is a 3D network of hydrophilic polymers with closely packed, linked tiny pores, allowing it to absorb large amounts of water rapidly **(Sadia et al., 2024)**. Their unique properties like biodegradability, biocompatibility also make SPHs have wide applications in various fields like biological, materials and pharmaceutical sciences **(Raghu Kiran & Gopinath, 2021)**. The superporous hydrogels have pores with dimensions of 50-100 m in size. These open pores are microscopic and are connected with each other, which allows them to absorb water rapidly by capillary action, increasing their size within one minute **(Mannan**

& Salim, 2025; Sadia et al., 2024). By adding hydrophilic materials like Ac-Di-Sol (Cross Carmellose Sodium), SPHs are able to grow to a greater size and develop a mechanical strength strong enough to withstand pressure from gastric contraction **(Lakhiew et al., 2026; Pawar et al., 2024)**.

Classification of superporous hydrogel

First generation as conventional superporous hydrogels(CSPHs)

Chen originally produced a highly porous hydrogel with remarkable absorption capacity and rapid swelling in 2000 **(Chen et al., 2000; Juthi et al., 2023)**. Vinyl monomer was used to prepare a conventional superporous hydrogel. The most commonly used monomers are those highly hydrophilic ones, such as acrylamide and sulfopropyl acrylate. Swollen SPHs are weak and difficult to handle due to their poor mechanical strength. Because of their weak nature, the formulation cannot remain in the stomach for too long time, as they readily disintegrate on slight pressure **(Issarachot et al., 2023; Juthi et al., 2023)**.

Second generation as superporous hydrogel composite(SPHCs)

This type of superporous hydrogel possesses a super disintegrant as a swellable filler. Baek (2001) has developed a second-generation superporous hydrogel from a normal superporous hydrogel**. (Farag et al., 2022)**. SPHs are prepared by using the composite ingredients like AC-Di-Sol, Primojel and crospovidone. The mechanical characteristics and swelling rate of AC-Di-Sol are both very high **(Farag et al., 2022; Munde et al., 2025)**. Acidification of the polymer ionizable groups is reported to improve the mechanical properties of the SPHs allowing the material to resist the strain of a stomach contraction in the gastric motility phase. The mechanical properties of the hydrogel composites will be improved by the composite agent **(Khushi & Vijeta, 2025; Pawar et al., 2024)**.

Third generation as superporous hydrogel hybrids(SPHHs)

Omidian created a hybrid superporous hydrogel comprised of acrylamide, methylene bisacryl amide and a cross linker in 2003 . Superporous hydrogel hybrids are known to have very high mechanical or elastic properties **(Sadia et al., 2024)**. SPHHs have a pre-cross-linked matrix swelling additive unlike superporous hydrogel composites. The hybrid agent like pectin and chitosan is a water-soluble polymer **(Vildanova et al., 2023)**. An example of an SPHH is where a super-porous hydrogel made of an acrylamide-based material in the presence of sodium alginate, then made into a hybrid by cross linking the sodium alginate with calcium ions. The water-swollen elastic superporous hydrogel hybrids are resistant to compressive, stiffness, twisting and bending **(Lavate et al., 2025)**.

Gastric Retention Mechanism of Superporous Hydrogel Systems

The SPHs retention in the stomach was mainly due to its rapid swelling property. When SPH was administered orally, the capacity of the small capsule was consumed within a few seconds, after that, it instantly swells to a large volume in the gastric fluid, making it restricting to pass into the intestine **(Al Shawakri et al., 2025; Grosso et al., 2023)**. The stomach muscle movements cause the stomach tissues to slide over the hydrogel. It can tolerate these movements due to its mechanical strength and its flexibility and slippery feature. It is also buoyant therefore it delivers drug in the upper GIT. The release of drug from the hydrogel breaks down slowly in the stomach. The drug is released slowly due to mechanical forces or the chemical enzymatic cleavage of the polymer chains in the hydrogel**(Al Shawakri et al., 2025; Díaz-Carrasco et al., 2025; Grosso et al., 2023)**.

The phases of gastric retention by superporous hydrogels are depicted in **Figure 4**, highlighting their swift expansion in gastric fluid and extended residence in the gastric region.

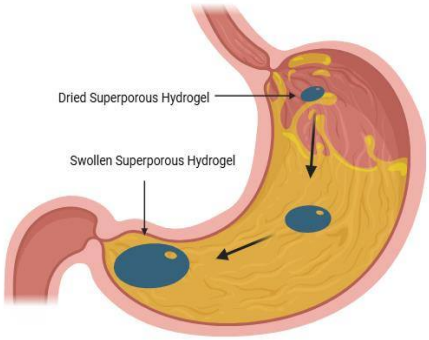


Figure 2: A representation of how superporous hydrogels are retained in the stomach after swelling there

Therapeutic Role in Peptic Ulcer Disease

The superporous hydrogel system was first developed for a regulated medication release mechanism within the gastric region. This system was developed to retain drug in the gastric region. The nature of the hydrogels system is spongy which is very important for application where the swelling of a particle needs to be very rapid. Superporous hydrogel systems have excellent retention in the stomach which make superporous hydrogel systems more efficient for the end user **(Choudhary et al., 2025)**. For treating ulcers, SPHs serve as carriers for the controlled and extended release of medications. Their large pore size and quick absorption allow for effective drug loading and release. Because of their fast water absorption and high expansion capacity, researchers are looking into SPHs for controlled-release and targeted drug delivery

(Issarachot et al., 2023; Raghu Kiran & Gopinath, 2021). These results imply that superporous hydrogel hybrid systems, which offer superior viscoelastic qualities and gastroretentive sustained administration of drugs with short residence durations, may be a viable choice **(Farag et al., 2022; Juthi et al., 2023; Sadia et al., 2024)**.

Challenges and Limitations of Superporous Hydrogels

Superporous hydrogels (SPHs) have great promise for targeted drug delivery that stays in the stomach. This is especially true for treating peptic ulcers. However, they face challenges. SPHs are prone to damage, show inconsistent behaviour in the body, and have formulation problems that affect how well they can hold and release medication **(Khan et al., 2022; Mustafa et al., 2024)**. In the stomach, the hydrogels take more time to achieve a state of swelling balance. The alkaline conditions enhance swelling efficiency for pH-sensitive hydrogels that include acidic monomers **(Khan et al., 2022; Mustafa et al., 2024)**. At a pH of 7.4, the hydrogel's water content significantly increased due to the presence of charged acidic carboxylic groups. The hydrogels constricted at lower pH levels due to the action of unprotonated carboxylic groups and expanded at higher pH levels due to the ionic repulsion from the protonated carboxylic groups. The ionized COO^- groups transformed into COOH groups when the buffer solution's pH fell to 1.2. This neutralized the ionic groups and made the hydrogels solid **(Garcia Ramos, 2025; Ullah et al., 2025)**.

Future prospects

The remarkable characteristics of SPHHs-enhanced elasticity, pH sensitivity, and excellent swelling capacity- warrant exploration for more extensive drug delivery applications down the line. Research will be conducted on pectin-based SPHHs for various pharmaceutical, biomedical, and industrial uses. Consequently, these polymers may serve as innovative carriers for the controlled-release of drugs. In vivo experiments demonstrated the hydrogel’s potential for promoting ulcer healing, indicating its possible application in the management of peptic ulcers. However, these results are initial and require further confirmation through comprehensive pharmacokinetic investigations, extended safety evaluations, and extensive in vivo assessments to establish the formulation's clinical applicability.

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